

## P1. Sales by Region

### Business Question:

Which region contributes the most to total sales revenue?

### What I Did:

Created a Pivot Table using **Region** as rows and aggregated **Sum of Sales** to compare revenue contribution across regions.

### Observation:

- **West** generated the highest sales revenue at **₹7,39,813.61**
- **East** followed closely with **₹6,91,828.17**
- **Central** contributed **₹5,03,170.67** in sales
- **South** generated the lowest sales revenue at **₹3,91,721.91**

### Insight:

Sales performance varies significantly across regions, indicating differences in customer demand, market penetration, product preference, or operational reach. The West region appears to be the strongest market, while South may represent an underperforming growth opportunity.

### Business Meaning / Recommendation:

The business should strengthen customer retention, inventory planning, and marketing investment in high-performing regions like **West** to maximize returns. At the same time, lower-performing regions such as **South** should be investigated to identify causes of weaker performance and uncover growth opportunities.

## P2. Profit by Category

### Business Question:

Which product categories contribute the most to total profit?

### What I Did:

Created a Pivot Table using **Category** as rows and aggregated **Sum of Profit** to compare profitability across product categories.

### Observation:

- **Technology** generated the highest total profit at **₹1,46,543.38**
- **Office Supplies** followed closely with **₹1,26,023.44**
- **Furniture** generated only **₹19,730.00**, significantly lower than the other categories

**Insight:**

Profitability differs substantially across categories. Although categories may contribute sales, they do not convert revenue into profit equally. Technology and Office Supplies demonstrate healthier margin performance, while Furniture shows signs of weak profitability.

**Business Meaning / Recommendation:**

The business should prioritize high-margin categories like **Technology** and **Office Supplies** through better promotion, inventory planning, and product expansion. Since **Furniture** contributes weak profit, management should review pricing, discounts, product mix, and operational costs to improve margin performance instead of focusing only on revenue growth.

## P3. Monthly Sales Trend

**Business Question:**

How have monthly sales changed over time?

**What I Did:**

Created a Pivot Table by grouping **Order Date into Month and Year** and aggregated **Sum of Sales** to analyze monthly sales performance over time. A line chart was used to visualize sales fluctuations and long-term trends.

**Observation:**

- Monthly sales fluctuate significantly across the timeline rather than increasing consistently.
- Several months show noticeable spikes in sales, particularly during later periods, indicating stronger business performance over time.
- Some months experience sharp declines after peak performance, showing uneven sales behavior.
- Overall, sales display an upward long-term trend despite short-term fluctuations.

**Insight:**

Sales performance appears to be influenced by seasonality, customer demand cycles, promotions, or product mix changes. The stronger peaks in later periods may indicate improving market presence, better customer acquisition, stronger repeat purchasing behavior, or operational growth over time.

**Business Meaning / Recommendation:**

The business should identify patterns behind high-performing months and replicate successful strategies such as promotions, inventory planning, product focus, or marketing campaigns. Since sales fluctuate noticeably, forecasting and seasonal

planning become important to reduce volatility and maintain consistent business growth.

## P4. Monthly Profit Trend

### **Business Question:**

How has monthly profit changed over time?

### **What I Did:**

Created a Pivot Table by grouping **Order Date into Month and Year** and aggregated **Sum of Profit** to analyze profitability trends over time. A line chart was used to visualize monthly profit fluctuations and long-term movement.

### **Observation:**

- Monthly profit fluctuates significantly across the timeline and does not follow a stable upward trend.
- Some months generated very strong profit, while a few periods experienced extremely low or even negative profit.
- Profit volatility appears much higher than sales volatility, showing inconsistent margin performance across months.
- Later periods generally show stronger profit spikes, indicating improvement in overall profitability over time.

### **Insight:**

Profit fluctuations suggest that strong sales alone are not consistently translating into stable profitability. Variations may be driven by changes in discounting, product mix, operational costs, seasonal demand, or margin efficiency. The stronger profit peaks in later periods may indicate better cost control, pricing optimization, or improved business maturity.

### **Business Meaning / Recommendation:**

The business should investigate what drives high-profit months and replicate successful pricing, product mix, and operational strategies. Since some periods generate weak or negative profit despite business activity, leadership should focus on margin stability rather than only revenue growth. Monitoring discounting, costs, and profitability trends can help reduce volatility and improve long-term business performance.

## P5. Top Customers by Sales

### Business Question:

Which customers contribute the most to total sales revenue?

### What I Did:

Created a Pivot Table using **Customer Name** as rows and aggregated **Sum of Sales**. The results were sorted in descending order to identify the highest revenue-generating customers.

### Observation:

- **Sean Miller** generated the highest sales revenue at **₹25,043.05**
- **Tamara Chand** contributed **₹19,052.22** in sales
- **Raymond Buch, Tom Ashbrook, and Adrian Barton** also contributed significant revenue with sales above **₹14,000** each
- A small group of customers contributes a disproportionately high share of revenue.

### Insight:

Revenue contribution is concentrated among a limited number of high-value customers, indicating that some customers have significantly stronger purchase behavior or larger transaction volumes than others.

### Business Meaning / Recommendation:

The business should prioritize retention of high-value customers through loyalty programs, personalized offers, stronger customer engagement, and premium service. However, customer value should be reviewed alongside profitability to ensure these customers generate sustainable business profit and not just high sales.

## P6. Loss-Making Subcategories

### Business Question:

Which product subcategories generate losses despite sales activity?

### What I Did:

Created a Pivot Table using **Subcategory Name** as rows and aggregated **Sum of Profit**. Applied filtering to identify only subcategories with **negative profit** to detect profit leakage areas.

### Observation:

- **Tables** generated the highest loss at **-₹17,753.21**
- **Bookcases** generated losses of **-₹3,632.07**

- **Supplies** recorded losses of **₹1,171.39**
- Losses are concentrated within a small number of subcategories rather than across all products.

#### **Insight:**

Certain subcategories generate revenue but fail to convert sales into profit, likely due to weak pricing, excessive discounting, low margins, high logistics costs, or inefficient product mix. The significant loss in **Tables** suggests a major profit leakage area.

#### **Business Meaning / Recommendation:**

The business should review pricing, discount policies, supplier costs, and profitability of loss-making subcategories — especially **Tables**. Rather than increasing sales volume, management should focus on fixing margin problems or reconsidering product strategy for underperforming subcategories to improve profitability without sacrificing revenue.

## **P7. Discount vs Profit Analysis**

#### **Business Question:**

How does discounting affect overall profitability?

#### **What I Did:**

Created a Pivot Table using **Discount percentage** as rows and aggregated **Sum of Profit** to analyze how profit changes across different discount levels. A chart was used to visualize the relationship between discounting and profitability.

#### **Observation:**

- Orders with **0% discount** generated the highest total profit at **₹3,26,718.59**
- Moderate discounts such as **10% and 20%** still generated positive profit, contributing **₹9,099.97** and **₹91,079.95** respectively
- Profit begins declining significantly at higher discount levels
- Discounts above **30%** mostly generated **negative profit**, with **70% and 80% discounts** causing the largest losses (**₹40,300.57** and **₹30,565.27**)
- Higher discounts do not consistently translate into better profitability.

#### **Insight:**

Heavy discounting appears to reduce margin efficiency and may convert revenue-generating sales into loss-making transactions. While small discounts may support customer acquisition or sales growth, aggressive discounting significantly weakens profitability.

#### **Business Meaning / Recommendation:**

The business should avoid excessive discounting and carefully evaluate whether

higher discounts are actually creating long-term business value. Instead of relying on deep discounts to drive sales, management should focus on pricing optimization, better product mix, bundling strategies, and targeted promotions that improve both revenue and profit. High discount brackets should be reviewed urgently to reduce profit leakage and protect margins.

## P8. Segment Analysis

### Business Question:

How do different customer segments contribute to sales and profit?

### What I Did:

Created a Pivot Table using **Segment** as rows and aggregated both **Sum of Sales** and **Sum of Profit** to compare performance across customer segments. A clustered column chart was used to visualize differences in sales and profitability.

### Observation:

- **Consumer** segment generated the highest sales (**₹11,70,659.79**) and highest profit (**₹1,36,371.45**)
- **Corporate** segment contributed **₹7,15,806.13** in sales and **₹94,249.64** in profit
- **Home Office** generated the lowest sales (**₹4,40,068.43**) and profit (**₹61,675.73**)
- Sales and profit appear to move together across segments, meaning stronger sales segments also contribute more profit.

### Insight:

The **Consumer segment** is the primary business driver, contributing the largest share of both revenue and profit. Meanwhile, Corporate and Home Office segments represent smaller but still meaningful revenue streams that may offer untapped growth potential.

### Business Meaning / Recommendation:

The business should strengthen retention and marketing efforts in the **Consumer segment** to maximize value while also exploring targeted strategies for **Corporate** and **Home Office** customers. Personalized offerings, segment-specific promotions, and business-focused packages could help improve performance in lower-contributing segments.

## P9. Sales by Ship Mode

### Business Question:

Which shipping mode contributes the most to total sales?

### What I Did:

Created a Pivot Table using **Ship Mode** as rows and aggregated **Sum of Sales** to compare revenue contribution across delivery methods.

### Observation:

- **Standard Class** generated the highest sales at **₹13,78,840.55**
- **Second Class** contributed **₹4,66,671.11** in sales
- **First Class** generated **₹3,51,750.74** in sales
- **Same Day** contributed the lowest sales at **₹1,29,271.96**

### Insight:

Customers appear to prefer cost-effective delivery methods over premium or urgent shipping options. **Standard Class** dominates sales contribution, suggesting it aligns best with customer expectations and purchasing behavior.

### Business Meaning / Recommendation:

The business should optimize operational efficiency and delivery quality for **Standard Class**, since it drives the majority of revenue. At the same time, premium shipping methods such as **First Class** and **Same Day** can be evaluated for pricing, customer targeting, or promotional opportunities to improve adoption without increasing logistics costs unnecessarily.

## P10. Returns Analysis

### Business Question:

How many customer orders were returned?

### What I Did:

Created a Pivot Table using **Returned status** and counted **Order IDs** to measure total returned orders.

### Observation:

A total of **296 orders** were returned by customers.

### Insight:

Returned orders represent operational inefficiency and potential profit leakage, as returns increase logistics costs, reduce realized revenue, and may signal customer dissatisfaction, product quality issues, or delivery-related problems.

**Business Meaning / Recommendation:**

The business should investigate the root causes of returns by analyzing affected products, customer segments, shipping methods, and regions. Reducing returns through better product quality, accurate product descriptions, improved delivery experience, and customer support can help protect both profitability and customer satisfaction.